

Assignment-2

Linear and Logistic Regression

Max Marks: 100

Due Date: 19th Feb 2026

Instructions:

- Assignments should be done on an **individual basis**.
- **Deliverables:** Well-documented code with generated output in `.ipynb` (or `.py`) and a PDF report file. Please ensure you submit all individual files (**Do Not submit in a .zip format**).

Part 1: Fish Market Regression (50 Points)

Data Source: `fish.xls` (provided)

Q1. Linear Regression (15 points) Fit a predictive linear regression model to estimate the **weight** of the fish from its length (`length1`, `length2`, `length3`), `height`, and `width`. Use only **70%** of the data as the training set.

- Report the coefficient values by using the standard Least Square Estimates.
- What is the standard error of the estimated coefficients, R-squared term, and the 95% confidence interval?
- Is there any dependence between the length and weight of the fish?

Q2. Ridge and Lasso Regression (15 points) Using the data source in Q1 (using only the same **70%** training set), fit the Ridge and Lasso Regression Models.

- Report the coefficients for both models.
- Report the attribute(s) **least impacting** the weight of the fish. Justify your answer.

Q3. Model Testing (20 points) Using the remaining **30%** of the data from Q1, test the trained regression models from Q1 and Q2.

- Report their performance (model accuracy) using **MSE** (Mean Squared Error) and **R²** metrics.
- Which model is performing the best? Explain your answer.

Part 2: Social Network Logistic Regression (50 Points)

Problem Statement: We have a dataset having 5 columns namely: **User ID**, **Gender**, **Age**, **EstimatedSalary**, and **Purchased**. We will build a Logistic Regression model that can predict whether, based on the given parameters, a person will buy a car or not.

Data Source: Please download the dataset from the link below:

<https://gist.github.com/davewadestein/5339c464e95f54f724f6f5381f0d95e1>

(a) Age and Salary Model (25 points) Please train and test the logistic regression model using **only Age and Estimated Salary** attributes.

- Use **70%** of the data for training and **30%** for testing.
- Report **F-1 score**, **Precision**, **Recall**, **Accuracy**, and the **ROC curve**.

(b) Gender, Age, and Salary Model (25 points) Please train and test the logistic regression model using **Gender, Age, and Estimated Salary** attributes.

- Use **70%** of the data for training and **30%** for testing.
- Report **F-1 score**, **Precision**, **Recall**, **Accuracy**, and the **ROC curve** for males and females, **separately**.
- Please report any performance differential across gender.

User ID	Gender	Age	EstimatedSalary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0
15728773	Male	27	58000	0
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	0
15727311	Female	35	65000	0
15570769	Female	26	80000	0
15606274	Female	26	52000	0
15746139	Male	20	86000	0
15704987	Male	32	18000	0